

Extended Derivations for “Automation, Capital Taxation, and Public Insurance”

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Contents

E1	Extended derivations in original economic coordinates	2
E2	The global firm problem and the maintained aggregate branch	3
E3	States, ownership, and the Ramsey problem	4
E3.1	Exogenous technology states	5
E3.2	Domestic capital and taxes	5
E3.3	External prices	6
E3.4	Public resources and admissibility	6
E4	The interior HJB and control recovery	7
E4.1	Five-state normalization	8
E5	Market and fiscal prices of risk	10
E6	Deterministic fiscal wealth and capital taxation	11
E6.1	Forward-looking tax adjustment	12
E6.2	The conditional fixed point and the Laffer comparison	12
E7	Automation incidence and transition results	13
E8	Small risk, portfolio finance, and incomplete spanning	14
E9	Commitment, recursion, and fiscal solvency	16
E10	The local quadratic benchmark	18
E11	Formal scope	20

Purpose and reading route

This supplement gives a continuous derivation in the model's original economic coordinates. The claim-organized Theory Appendix is the shorter route for checking a particular proposition. This volume is for readers who instead want to follow the whole chain from the firm's task allocation and the decentralized accounts to fiscal and market prices of risk, the tax condition, the small-risk expansion, commitment, and the local quadratic system.

The first three numbered sections build the firm problem, ownership, and Ramsey continuation in levels. The next four derive the traded-direction condition, deterministic fiscal wealth, automation incidence, and incomplete spanning. The final sections record commitment, solvency, and the local quadratic representation. Solver algorithms and numerical acceptance belong to the Computation Appendix.

The same scope applies throughout: the derivations characterize the stated smooth branch and preserve each proposition's hypotheses. They do not establish a global solution of the constrained five-state problem. Capital K , public financial net worth N , worker consumption c , the dollar risky holding s , and the tax rate τ appear before normalized or matrix objects.

E1 Extended derivations in original economic coordinates

This module keeps the complete derivation in levels and introduces normalized coordinates only when they become useful. It complements the claim-organized modules above: those modules let a reader audit a particular result quickly, whereas the derivation below lets a reader follow the whole chain from the firm problem to the local system without reconstructing omitted algebra.

The main paper is organized around an economic comparison: international markets and a worker-welfare government value the same shock differently, and the public portfolio eliminates that difference only along the payoff that can be traded. This appendix develops the argument from the primitives. Its aim is not merely to collect proofs. It explains how ownership, timing, and the public budget enter the planner's problem; how the portfolio first-order condition becomes a comparison of prices of risk; and which parts of the argument rely on interiority, small risk, or a local approximation.

The argument has three layers. Sections E2 and E3 define the economic environment. They show how the task economy reduces to the production function used in the paper, who owns capital and financial assets, and which choices are made before and after a shock. Sections E4–E8 derive the economic results: the exact interior controls, the market–fiscal risk-price equality, the deterministic fiscal-wealth benchmark, the tax condition, and the cost of incomplete spanning. Sections E9 and E10 then explain why the problem can be written recursively and how the local benchmark is obtained.

The notation moves from economic levels to solution coordinates. Capital K , public net worth

N , the risky holding s , worker consumption c , and the tax rate τ come first. Lower-case ratios such as $n = N/K$ and $\theta = s/K$ appear only when the problem is normalized for computation. A risky holding is a dollar position; a portfolio share is that position divided by an explicitly named wealth measure.

The formal statements below hold under the mathematical conditions stated with each result. They characterize a smooth, nondegenerate interior branch. They are not a global existence or verification theorem for the constrained five-state problem.

E2 The global firm problem and the maintained aggregate branch

The aggregate production function in the paper is valid when capital performs the automation-intensive activities and labour performs the new tasks. The global firm problem is needed here to show when that allocation is optimal and what happens at its boundaries.

At each date, the firm assigns capital and labour between an automation-intensive composite A and a new-task composite M . The letter M avoids confusing the new-task input with public net worth, denoted N elsewhere. The composite inputs are

$$y_A = e^{\nu_A} K_A + L_A, \quad y_M = e^{-\nu_N} K_M + L_M, \quad \nu_A, \nu_N > 0. \quad (\text{E2.1})$$

Capital has comparative advantage in A and labour in M . For fixed (z, α, K, L) , output is

$$Y = \max_{\substack{K_A, K_M, L_A, L_M \geq 0 \\ K_A + K_M = K, L_A + L_M = L}} e^z \left(\frac{e^{\nu_A} K_A + L_A}{\alpha} \right)^\alpha \left(\frac{e^{-\nu_N} K_M + L_M}{1 - \alpha} \right)^{1-\alpha}. \quad (\text{E2.2})$$

Define

$$\Omega(\alpha) = \alpha^{-\alpha} (1 - \alpha)^{-(1-\alpha)}, \quad (\text{E2.3})$$

and the specialization margins

$$s_A = \log(K/L) - \text{logit}(\alpha) + \nu_A, \quad s_N = \nu_N - \log(K/L) + \text{logit}(\alpha). \quad (\text{E2.4})$$

Proposition E2.1 (Global task allocation). *The value of (E2.2) is*

$$Y = \begin{cases} e^z (e^{\nu_A} K + L), & s_A \leq 0, \\ e^{z + \nu_A \alpha} \Omega(\alpha) K^\alpha L^{1-\alpha}, & s_A \geq 0, s_N \geq 0, \\ e^{z + \nu_A \alpha - \nu_N (1-\alpha)} (K + e^{\nu_N} L), & s_N \leq 0. \end{cases} \quad (\text{E2.5})$$

Adjacent formulas agree at a boundary. The middle branch assigns $K_A = K$ and $L_M = L$. It is strict if and only if $s_A > 0$ and $s_N > 0$.

The three cases describe scarcity, specialization, and abundance of capital. When capital is scarce, some labour also enters the automation-intensive composite. In the middle region the comparative-advantage allocation is complete: capital performs A and labour performs M . When capital is abundant, some of it spills into the new-task composite. The first margin prevents labour from entering A and is the condition that all available automation-intensive tasks are privately performed with capital. The second prevents capital from entering M . Full adoption in the first sense therefore does not by itself justify the smooth middle formula.

On the middle branch,

$$R^K = \alpha Y/K, \quad w = (1 - \alpha)Y/L, \quad (\text{E2.6})$$

and the two inequalities are equivalent to

$$\frac{R^K}{e^{\nu_A}} < w < e^{\nu_N} R^K. \quad (\text{E2.7})$$

The rental flow R^K is a producer price per unit of installed capital, not a financial interest rate.

Holding (z, K, L) fixed, the optimized output derivative is

$$\frac{\partial \log Y}{\partial \alpha} = \begin{cases} 0, & s_A < 0, \\ s_A, & s_A > 0, \ s_N > 0, \\ \nu_A + \nu_N, & s_N < 0. \end{cases} \quad (\text{E2.8})$$

Optimized output is therefore globally weakly increasing in available automation. It is flat in the capital-scarce regime because the firm can reproduce the same effective input by reallocating labour. It rises in the middle and capital-abundant regions because the expanded automation set raises the productivity of the allocation the firm already wants to use.

For a rectangle $k = \log(K/L) \in [k_{\min}, k_{\max}]$ and $\alpha \in [\alpha_{\min}, \alpha_{\max}]$, the middle branch is strict throughout if

$$k_{\min} > -\nu_A + \text{logit}(\alpha_{\max}), \quad k_{\max} < \nu_N + \text{logit}(\alpha_{\min}). \quad (\text{E2.9})$$

The minimum of the two left-hand gaps is a uniform specialization margin. This is the useful numerical form of the result: a proposed domain can be compared with the two thresholds before the smooth production formula is used there.

E3 States, ownership, and the Ramsey problem

The timing is simple but important. At the start of an instant, productivity, automation, installed capital, public net worth, and the current tax rate are inherited. The risky position chosen just before the innovation determines the contemporaneous gain or loss on the public balance sheet. Capital and the tax rate cannot jump. After the innovation, the government chooses worker

consumption, the next risky position, and the speed at which the tax rate will change. This distinction is why financial insurance and capital taxation are not the same instrument.

E3.1 Exogenous technology states

Productivity and the latent automation state follow independent Ornstein–Uhlenbeck processes,

$$dz = \kappa_z(\bar{z} - z) dt + \sigma_z dW^P, \quad (\text{E3.1})$$

$$dx = \kappa_x(\bar{x} - x) dt + \sigma_x dW^A, \quad d\langle W^P, W^A \rangle_t = 0. \quad (\text{E3.2})$$

We collect the innovations in the order productivity, automation:

$$d\mathbf{W} = \begin{pmatrix} dW^P \\ dW^A \end{pmatrix}. \quad (\text{E3.3})$$

The available automation share is the bounded increasing function

$$\alpha = A(x) = \underline{\alpha} + \frac{\bar{\alpha} - \underline{\alpha}}{1 + e^{-x}}. \quad (\text{E3.4})$$

Write $a(x) = A'(x) > 0$ and $q(x) = A''(x)$.

E3.2 Domestic capital and taxes

All domestic installed capital is privately owned. The government does not own K , and public net worth N does not include the domestic capital stock. Its installed price is normalized to one and installation is reversible one for one. Domestic capitalists have log utility, consume $C^K = \rho K$, and are segmented from the external asset market. Their saving decision closes domestic capital accumulation even though their utility is not included in the planner's stated objective.

The source-based tax applies to net rental income,

$$\mathcal{B} = (R^K - \delta)K. \quad (\text{E3.5})$$

which may be negative when $R^K < \delta$. The tax rate is inherited and finite variation,

$$d\tau = \nu dt, \quad \tau \leq 1, \quad (\text{E3.6})$$

and adjustment costs are $Y\nu^2/(2\kappa_\tau)$. This smooth reduced form makes the tax rate a policy state: it summarizes legislative, implementation, and private transition costs that prevent costless state-by-state resetting. With no adjustment friction, the rate could respond immediately to the current state for subsequent rental flows. It would still not pay on the Brownian innovation already realized. Here the level of the rate cannot move on impact, so the pre-positioned financial claim

moves public net worth with the innovation and tax policy changes afterward. Alternative costs or implementation lags can change the speed and persistence of the continuation response.

Capital obeys

$$dK = Kg \, dt, \quad g = (1 - \tau)(R^K - \delta) - \rho. \quad (\text{E3.7})$$

Thus $[K]_t = [K, W]_t = 0$. A technology shock changes the return earned by installed capital and hence its subsequent growth rate, but does not instantaneously create or destroy installed units.

E3.3 External prices

The government can save or borrow at the international safe rate and trade one risky claim. These prices are taken as given by the small domestic economy. For a general external claim, let σ_I be its return loading and λ_I the international market price of risk. No arbitrage gives

$$\beta_I = \sigma_I^\top \lambda_I, \quad d\mathcal{R}_I = (r_I + \beta_I) \, dt + \sigma_I^\top d\mathbf{W}. \quad (\text{E3.8})$$

The maintained external closure specializes further. International capital is fixed, log utility prices the aggregate risky claim, and aggregate external safe supply is zero. If $k_I = \log(\bar{K}_I/L_I)$ and $\ell_I = \log C_I^K$, then

$$\ell_{I,z} = 1, \quad \ell_{I,x} = a(x)h_I(\alpha), \quad h_I(\alpha) = \frac{1}{\alpha} + k_I - \text{logit}(\alpha) + \nu_A. \quad (\text{E3.9})$$

Consequently,

$$\sigma_I = \lambda_I = \begin{pmatrix} \sigma_z \\ a(x)h_I(\alpha)\sigma_x \end{pmatrix}, \quad \beta_I = \|\lambda_I\|^2, \quad (\text{E3.10})$$

and

$$r_I = \rho + \mathcal{L}^{z,x}\ell_I - \frac{1}{2}\beta_I. \quad (\text{E3.11})$$

The first component of the claim pays with productivity; the second pays with automation according to its effect on external capitalist consumption. The equality $\sigma_I = \lambda_I$ is specific to the log-utility external closure, not a general asset-pricing identity. The domestic economy is interpreted as infinitesimal, or a residual external sector absorbs its positions without moving prices. There are two primitive shocks but only one risky payoff, so the traded span has rank one.

E3.4 Public resources and admissibility

Let N be public net worth, s the market value of the risky claim, and c worker consumption. The residual $N - s$ is the safe asset position; it is negative when the government borrows. The wage bill is $\mathcal{W} = wL$, transfers are $T = c - \mathcal{W} \geq 0$, and consolidated worker-planner resources before

consumption are

$$\mathcal{F} = \tau\mathcal{B} + \mathcal{W}. \quad (\text{E3.12})$$

The public budget is

$$dN = \left[r_I N + s\beta_I + \mathcal{F} - c - \frac{Y}{2\kappa_\tau} \nu^2 \right] dt + s\sigma_I^\top d\mathbf{W}. \quad (\text{E3.13})$$

The drift of N contains the return on net worth, the risky premium, capital-tax revenue, transfers, and the resource cost of changing taxes. Its diffusion is the risky holding multiplied by the claim's payoff. Since $\mathcal{F} - c = \tau\mathcal{B} - T$, wages are not Treasury receipts. They appear in consolidated resources only because the planner values worker consumption directly.

Portfolio constraints take the form

$$-\bar{S}^G \leq s \leq N + \bar{B}^G. \quad (\text{E3.14})$$

The right inequality is a safe-debt limit because the safe position is $N - s$. The feasible interval is nonempty only if $N \geq -(\bar{B}^G + \bar{S}^G)$.

The planner values workers, who are hand to mouth, and uses domestic capitalists' behaviour as an equilibrium constraint. It solves

$$V(S_0) = \sup_{(c,s,\nu) \in \mathcal{A}^{ad}(S_0)} \mathbb{E}_{S_0} \int_0^\infty e^{-\rho t} \log c_t \, dt, \quad S = (z, x, K, N, \tau). \quad (\text{E3.15})$$

Admissibility includes positive consumption, the transfer floor, portfolio limits, the tax ceiling and inward tax speed there, continuation feasibility after every reached stopping time, integrability, and conditional no-Ponzi and transversality restrictions. These restrictions do not change the interior first-order conditions below. They determine where those conditions can be used and what must happen when a boundary is reached.

E4 The interior HJB and control recovery

The value function $V(z, x, K, N, \tau)$ records the utility attainable from the current economic state. Its derivative V_N is the marginal value of one more unit of public net worth. Concavity, $V_{NN} < 0$, says that this marginal value falls as the public balance sheet improves. The derivatives V_K and V_τ measure the continuation value of installed capital and of inheriting a different tax rate.

For a smooth test function ϕ , define

$$D[\phi] = \sigma_z \sigma_I^P \phi_{zN} + \sigma_x \sigma_I^A \phi_{xN}. \quad (\text{E4.1})$$

This covariance term appears because the exogenous states and public net worth share the same

Brownian shocks. Buying the claim therefore changes not only the variance of N but also its covariance with the technology states. The pre-optimization HJB is

$$\begin{aligned} \rho V = \max_{c,s,\nu} & \left\{ \log c + \mathcal{L}^{z,x} V + KgV_K + \nu V_\tau \right. \\ & + \left[r_I N + s\beta_I + \mathcal{F} - c - \frac{Y}{2\kappa_\tau} \nu^2 \right] V_N \\ & \left. + sD[V] + \frac{s^2 \|\boldsymbol{\sigma}_I\|^2}{2} V_{NN} \right\}. \end{aligned} \quad (\text{E4.2})$$

The first line contains current utility and the continuation effects of technology, capital accumulation, and tax adjustment. The second values the expected change in public net worth. The final line contains the covariance and variance effects of the risky position. Here $\beta_I = \boldsymbol{\sigma}_I^\top \boldsymbol{\lambda}_I$. If $V_N > 0$, $V_{NN} < 0$, and the controls are interior, the pointwise maximizers are

$$c^* = \frac{1}{V_N}, \quad \nu^* = \frac{\kappa_\tau}{Y} \frac{V_\tau}{V_N}, \quad s^* = -\frac{\beta_I V_N + D[V]}{\|\boldsymbol{\sigma}_I\|^2 V_{NN}}. \quad (\text{E4.3})$$

The consumption rule equates the marginal utility of worker consumption to the marginal value of retaining a public dollar. Tax speed is high when inheriting a larger tax rate has high continuation value and low when adjustment is costly. The portfolio balances its expected excess return against both the curvature of the value function and the way its payoff covaries with the other states.

Substitution gives

$$\begin{aligned} \rho V = & -\log V_N - 1 + \mathcal{L}^{z,x} V + KgV_K + (r_I N + \mathcal{F}) V_N \\ & + \frac{\kappa_\tau}{2Y} \frac{V_\tau^2}{V_N} - \frac{(\beta_I V_N + D[V])^2}{2\|\boldsymbol{\sigma}_I\|^2 V_{NN}}. \end{aligned} \quad (\text{E4.4})$$

Equation (E4.4) is the nonlinear interior equation to be solved. Its last two terms are the gains from optimally choosing tax speed and the risky position. They are positive because the maximizations improve on setting those controls to zero.

At the transfer floor, $c^* = \max\{\mathcal{W}, 1/V_N\}$. At a portfolio bound, the scalar maximizer in (E4.3) is projected onto (E3.14). At $\tau = 1$, tax speed is projected onto $\nu \leq 0$. Continuation solvency and production switches are state or regime conditions rather than pointwise projections. Thus the constrained problem is not obtained by using the interior formula everywhere; each boundary changes the relevant local choice or the admissible continuation set.

E4.1 Five-state normalization

The level problem is homogeneous in capital and public financial wealth. Dividing flow variables and financial positions by K removes this scale without changing their economic meaning. The

transformed state $k = \log(K/L)$ retains capital abundance relative to labour, while $n = N/K$ records the public balance sheet relative to the domestic capital stock.

Define

$$k = \log(K/L), \quad n = N/K, \quad \tilde{c} = c/K, \quad \theta = s/K, \quad (\text{E4.5})$$

and, on the full-specialization branch,

$$y = Y/K, \quad w = \mathcal{W}/K = (1 - \alpha)y, \quad b = \mathcal{B}/K = \alpha y - \delta, \quad f = \tau b + w. \quad (\text{E4.6})$$

Here $\tilde{c}$ is worker consumption per unit of capital and θ is the risky holding per unit of capital; neither is a portfolio share. Since K has finite variation,

$$dn = \left[(r_I - g)n + \theta\beta_I + f - \tilde{c} - \frac{y}{2\kappa_\tau}\nu^2 \right] dt + \theta\lambda_I^\top dW \quad (\text{E4.7})$$

under the specialized claim closure.

Public net worth per unit of capital rises with asset returns and consolidated resources, and falls with consumption, adjustment costs, and capital growth. Write the normalized value as

$$U(z, x, k, n, \tau) = \frac{\log L + k}{\rho} + v(z, x, k, n, \tau). \quad (\text{E4.8})$$

With

$$D_v = \sigma_z \lambda_I^P v_{zn} + \sigma_x \lambda_I^A v_{xn}, \quad (\text{E4.9})$$

the interior PDE becomes

$$\begin{aligned} \rho v = & -\log v_n - 1 + \mathcal{L}^{z,x}v + g \left(\frac{1}{\rho} + v_k \right) + [(r_I - g)n + f]v_n \\ & + \frac{\kappa_\tau}{2y} \frac{v_\tau^2}{v_n} - \frac{(\beta_I v_n + D_v)^2}{2\beta_I v_{nn}}. \end{aligned} \quad (\text{E4.10})$$

The feedback rules are

$$\tilde{c}^* = \frac{1}{v_n}, \quad \nu^* = \frac{\kappa_\tau}{y} \frac{v_\tau}{v_n}, \quad \theta^* = -\frac{\beta_I v_n + D_v}{\beta_I v_{nn}}. \quad (\text{E4.11})$$

The normalized controls recover consumption, tax speed, and the risky holding in the same units. This is the current five-state computational target. It is valid only where $v_n > 0$, $v_{nn} < 0$, $\beta_I > 0$, and every maintained boundary is slack.

E5 Market and fiscal prices of risk

The portfolio condition can be read without solving the entire HJB. International investors price risk according to the exposure of their marginal utility. The government prices the same innovation according to the exposure of V_N , the marginal value of a public dollar. A marginal purchase is desirable when the claim is better compensated by markets than it is costly in fiscal-value terms.

Define the fiscal price of Brownian risk at a candidate holding s by

$$\lambda_G(s) = -\frac{1}{V_N} \begin{pmatrix} \sigma_z V_{Nz} + s\sigma_I^P V_{NN} \\ \sigma_x V_{Nx} + s\sigma_I^A V_{NN} \end{pmatrix}. \quad (\text{E5.1})$$

It is the negative diffusion of the log fiscal stochastic discount factor $e^{-\rho t} V_N$. On the transfer interior, $V_N = 1/c$, so λ_G is also the diffusion loading of log c .

The current holding enters $\lambda_G(s)$ because it changes how public net worth responds to each shock. Buying more of the claim can therefore change the government's own marginal valuation of the payoff. The derivative of the portfolio Hamiltonian is

$$\mathcal{H}_s = \beta_I V_N + D[V] + s \|\sigma_I\|^2 V_{NN}. \quad (\text{E5.2})$$

Its three terms are expected excess return valued at V_N , covariance with the other states, and the marginal variance cost from concavity. Using $\beta_I = \sigma_I^\top \lambda_I$ and (E5.1),

$$\frac{\mathcal{H}_s(s)}{V_N} = \sigma_I^\top [\lambda_I - \lambda_G(s)]. \quad (\text{E5.3})$$

Derivation of traded-direction orthogonality. Suppose transfers and the portfolio are interior. The first-order condition implies

$$\sigma_I^\top (\lambda_I - \lambda_G) = 0. \quad (\text{E5.4})$$

Under the specialized closure $\sigma_I = \lambda_I$, this is equivalent to

$$\frac{\text{Cov}_t \left(d \log(c_t / C_{I,t}^K), d\mathcal{R}_{I,t} \right)}{dt} = 0. \quad (\text{E5.5})$$

At an unconstrained optimum $\mathcal{H}_s = 0$, so (E5.3) yields (E5.4). Under interior transfers, λ_G is the diffusion of log c ; under the external log-owner closure, λ_I is the diffusion of both log C_I^K and the claim return. The covariance in (E5.5) is therefore $(\lambda_G - \lambda_I)^\top \lambda_I dt$.

The result concerns relative consumption along the traded return. It is not an absolute worker-consumption variance-minimization result and does not imply $\lambda_G = \lambda_I$. With two shocks and one claim, the equality imposes one scalar restriction: the difference between market and fiscal prices

may remain in the direction the claim does not pay.

If the upper portfolio bound binds, strict concavity gives $\mathcal{H}_s \geq 0$ and the covariance in (E5.5) is weakly negative. If the lower bound binds, $\mathcal{H}_s \leq 0$ and the covariance is weakly positive. The local shadow value of relaxing the active bound is the corresponding Hamiltonian residual.

The same first-order condition gives a familiar return-demand and hedging decomposition:

$$s^* = \underbrace{\frac{\beta_I}{\|\sigma_I\|^2} \left(-\frac{V_N}{V_{NN}} \right)}_{\text{return demand}} - \underbrace{\frac{D[V]}{\|\sigma_I\|^2 V_{NN}}}_{\text{fiscal hedge}}. \quad (\text{E5.6})$$

The return-demand term is expected excess return per unit of variance multiplied by fiscal risk tolerance. The hedge offsets the marketed component of the risk already contained in the continuation value of wages, tax receipts, capital, and future policy. Writing the dollar holding directly avoids confusing it with a portfolio share; a share is introduced below only after total planner wealth has been defined.

E6 Deterministic fiscal wealth and capital taxation

Removing instantaneous risk separates the intertemporal fiscal problem from the insurance problem. The government then has two kinds of resources: financial net worth already on its balance sheet and the present value of future wages and tax receipts it can generate while managing the tax rate. The latter is fiscal wealth.

Evaluate the deterministic benchmark at the common-state means, where the zero-risk external safe rate equals the common discount rate. Fiscal wealth is

$$J(K, \tau) = \sup_{\nu} \int_0^{\infty} e^{-\rho t} \left[\mathcal{F}(K_t, \tau_t) - \frac{Y(K_t)}{2\kappa_{\tau}} \nu_t^2 \right] dt, \quad (\text{E6.1})$$

subject to (E3.7) and $\dot{\tau} = \nu$. The flow inside the integral combines worker wages and capital-tax revenue, net of the cost of changing the tax rate. Because the planner values workers, wages belong in this resource measure even though they are not Treasury receipts. Its HJB is

$$\rho J = \mathcal{F} + KgJ_K + \frac{\kappa_{\tau}}{2Y} J_{\tau}^2, \quad \nu^* = \frac{\kappa_{\tau}}{Y} J_{\tau}. \quad (\text{E6.2})$$

Let $X = N + J$ denote total planner wealth. The equality is an economic decomposition, not a government balance-sheet identity: N is a financial asset position, whereas J is the value of future domestic resources. On a globally feasible interior characteristic,

$$V^0 = \frac{1}{\rho} \log(\rho X), \quad c^0 = \rho X. \quad (\text{E6.3})$$

The deterministic problem therefore has the same consumption rule as a log investor whose wealth

is X . With deterministic mean reversion away from the state means, an additive opportunity-value term may enter V^0 but does not change these derivatives with respect to N .

E6.1 Forward-looking tax adjustment

The current tax rate affects more than today's revenue. Raising it changes the subsequent capital stock and hence future wages and tax bases. The optimal adjustment rule compares the direct revenue gain with the fiscal value of the capital displaced by the tax.

Let $m = J_\tau$ and $\ell = J_K$. The derivative m is the value of inheriting a slightly higher tax rate; ℓ is the value of another unit of installed capital. The current-value tax equations are

$$\nu = \frac{\kappa_\tau}{Y} m, \quad \dot{m} = \rho m + \mathcal{B}(\ell - 1). \quad (\text{E6.4})$$

If $e^{-\rho(T-t)} m_T \rightarrow 0$, integration forward gives

$$J_\tau(K_t, \tau_t) = \int_t^\infty e^{-\rho(u-t)} \mathcal{B}_u [1 - J_K(K_u, \tau_u)] du, \quad (\text{E6.5})$$

and hence

$$\nu_t = \frac{\kappa_\tau}{Y_t} \int_t^\infty e^{-\rho(u-t)} \mathcal{B}_u [1 - J_K(K_u, \tau_u)] du. \quad (\text{E6.6})$$

The present-value formula makes the trade-off explicit. At each future date the net rental base $\mathcal{B}_u$ measures the direct resources from a marginally higher rate. The term $1 - J_K$ subtracts the fiscal value of preserving the capital that the tax discourages. Tax adjustment looks forward because a change made today alters this entire stream.

E6.2 The conditional fixed point and the Laffer comparison

At a smooth interior fixed point with $\mathcal{B}^* > 0$, tax speed and its costate vanish. The tax costate equation implies $J_K = 1$: the marginal fiscal value of installed capital equals its installed price. The capital and fiscal return conditions become

$$\rho = (1 - \tau)(R^K - \delta), \quad \rho = \tau(R^K - \delta). \quad (\text{E6.7})$$

Thus

$$\tau^* = \frac{1}{2}, \quad R^{K,*} - \delta = 2\rho. \quad (\text{E6.8})$$

The two equations assign the same net rental return to private accumulation and public fiscal wealth. Their equality produces the one-half result. This conclusion uses common discounting and an interior convergent path; changing those assumptions changes the comparison.

For a fixed tax rate, stationary capital is

$$K^*(\tau) = \left[\frac{(1-\tau)\alpha A}{\rho + (1-\tau)\delta} \right]^{1/(1-\alpha)}, \quad (\text{E6.9})$$

where A collects fixed productivity terms. Stationary capital-tax revenue is $\mathcal{T}(\tau) = \rho\tau K^*(\tau)/(1-\tau)$ and has its unique peak at

$$\tau_{\text{Laffer}} = \frac{(1-\alpha)(\rho + \delta)}{\rho + (1-\alpha)\delta}. \quad (\text{E6.10})$$

It equals one half only on a knife edge. The Ramsey fixed point is not a revenue-maximization result. It balances current fiscal resources against the worker-value of future capital, whereas the Laffer peak asks only which constant rate maximizes stationary tax receipts.

E7 Automation incidence and transition results

Automation changes both productive capacity and the division of income. The first set of derivatives holds installed capital and the inherited tax rate fixed, which is the relevant impact comparison when an innovation arrives. The second comparison allows capital to adjust across deterministic steady states.

On the full-specialization branch, differentiation with respect to the latent state x gives

$$Y_x = as_A Y > 0, \quad (\text{E7.1})$$

$$\mathcal{B}_x = aY(1 + \alpha s_A) > 0, \quad (\text{E7.2})$$

$$\mathcal{W}_x = aY[(1-\alpha)s_A - 1], \quad (\text{E7.3})$$

$$\mathcal{F}_x = aY[\tau - 1 + (1-\alpha + \tau\alpha)s_A], \quad (\text{E7.4})$$

$$g_x = (1-\tau)ah_D R^K, \quad (\text{E7.5})$$

where $h_D = 1/\alpha + s_A > 0$. Output rises because more activities can use capital's comparative advantage. The rental base rises for the same reason. The wage bill has an additional negative term because labour is displaced from the newly automated activities. Hence automation can lower wages and current fiscal resources while increasing output, the tax base, and capital growth.

The wage effect is negative exactly when

$$s_A < \frac{1}{1-\alpha}. \quad (\text{E7.6})$$

If $1 - \alpha + \tau\alpha > 0$, current consolidated resources rise if and only if

$$s_A > \frac{1-\tau}{1-\alpha + \tau\alpha}. \quad (\text{E7.7})$$

Holding the tax rate fixed across deterministic steady states, capital adjustment gives

$$\frac{d \log w^{ss}}{dx} = \frac{a(x)s_A^{ss}}{1-\alpha} > 0. \quad (\text{E7.8})$$

The coexistence of a negative impact wage response and a positive long-run response therefore requires $0 < s_A < 1/(1-\alpha)$ in the relevant region. In this case automation initially hurts workers but eventually raises their wage through the capital it induces.

Two local directions display the spanning problem. They are not proposed shock processes; they are thought experiments that combine productivity and automation innovations to hold one observable payoff fixed. At inherited capital, $d \log Y = dz + s_A d\alpha$. Along the output-neutral direction $dz = -s_A d\alpha$,

$$dY = 0, \quad dW = -Y d\alpha, \quad d\mathcal{B} = Y d\alpha, \quad d\mathcal{F} = -(1-\tau)Y d\alpha. \quad (\text{E7.9})$$

If domestic and external factor intensities are locally aligned, external capitalist consumption rises by $d\alpha/\alpha$ along this direction. The claim pays while current fiscal resources fall.

Along the claim-neutral direction $dz = -h_I d\alpha$, external capitalist consumption and the maintained claim have no innovation. Under the same alignment condition,

$$d \log \mathcal{W} = -\frac{d\alpha}{\alpha(1-\alpha)} < 0. \quad (\text{E7.10})$$

This direction is locally invisible to the claim despite a domestic wage loss. It is the sharpest local illustration of incomplete spanning: the government values insurance in a state for which the only risky claim delivers no payoff.

E8 Small risk, portfolio finance, and incomplete spanning

The exact portfolio rule depends on derivatives of the full nonlinear value function. The small-risk expansion supplies a transparent benchmark. It shrinks the volatility of both primitive shocks while preserving their relative size and the direction of the marketed payoff. The deterministic fiscal-wealth problem then provides the leading marginal values used in the portfolio condition.

Scale both primitive volatilities by ε :

$$\sigma_z = \varepsilon \hat{\sigma}_z, \quad \sigma_x = \varepsilon \hat{\sigma}_x. \quad (\text{E8.1})$$

Under the specialized external closure,

$$\lambda_I = \varepsilon \hat{\lambda}, \quad \hat{\lambda} = \begin{pmatrix} \hat{\sigma}_z \\ a(x)h_I(\alpha)\hat{\sigma}_x \end{pmatrix}, \quad \hat{\beta} = \|\hat{\lambda}\|^2. \quad (\text{E8.2})$$

Let $J(z, x, K, \tau)$ be deterministic fiscal wealth and let $X = N + J$ be total planner wealth. The exposure of fiscal wealth to the two standardized innovations is

$$\zeta_J = \begin{pmatrix} \hat{\sigma}_z J_z \\ \hat{\sigma}_x J_x \end{pmatrix}, \quad h_J = \frac{\zeta_J^\top \hat{\lambda}}{\hat{\beta}}. \quad (\text{E8.3})$$

The scalar h_J is the amount of the marketed claim that reproduces the projection of fiscal-wealth risk onto its payoff. If fiscal wealth rises in exactly the states in which the claim pays, h_J is positive and the government needs less of the claim for hedging.

Derivation of the leading risky and safe positions. Suppose $\hat{\beta} > 0$ and the leading position is interior. Then the algebraic desired risky portfolio share and safe position are

$$\frac{s^0}{X} = 1 - \frac{h_J}{X}, \quad N - s^0 = h_J - J. \quad (\text{E8.4})$$

Thus the algebraic desired safe position is negative exactly when $J - h_J > 0$, and the desired risky position is positive when $N + J - h_J > 0$. These sign statements describe unconstrained demands, not implementable borrowing capacity; portfolio, transfer, and continuation-solvency constraints may prevent those positions.

Substitute the deterministic derivatives $V_N^0 = 1/(\rho X)$ and $V_{NN}^0 = -1/(\rho X^2)$ into the portfolio first-order condition after cancelling the common factor ε^2 . The return-demand component becomes total planner wealth, and the covariance term becomes the projection in (E8.3). Hence $s^0 = X - h_J$, which yields both expressions in (E8.4).

The first expression is written as a share because this is the standard portfolio object: the risky dollar holding divided by the wealth against which risk is borne. Without fiscal endowment risk, the log benchmark invests all total planner wealth in the aggregate claim. The term h_J/X is the hedging correction.

On the separated branch, $\partial s^0/\partial N = 1$ and $\partial(N - s^0)/\partial N = 0$: additional inherited public net worth is invested one for one in the risky claim, while the desired safe position is pinned down by fiscal wealth and its marketed exposure. Any binding transfer, balance-sheet, or solvency constraint breaks this invariance.

The order- ε^2 welfare gain from access to the claim rather than the bond-only restriction is

$$V^{\text{claim}} - V^{\text{bond}} = \varepsilon^2 \int_0^\infty e^{-\rho t} \frac{\hat{\beta}_t}{2\rho} \left(\frac{s_t^0}{X_t} \right)^2 dt + O(\varepsilon^4), \quad (\text{E8.5})$$

when both positions are feasible along the common deterministic baseline. The integrand is larger when the marketed Sharpe ratio is high and when the desired risky share is far from zero. This is the value of access to the existing claim, not the value of completing markets.

Define the unspanned residual

$$\zeta_J^\perp = \zeta_J - \frac{\zeta_J^\top \hat{\lambda}}{\hat{\beta}} \hat{\lambda}. \quad (\text{E8.6})$$

With nonzero primitive volatilities,

$$\|\zeta_J^\perp\|^2 = \frac{\hat{\sigma}_z^2 \hat{\sigma}_x^2 [J_x - a(x) h_I(\alpha) J_z]^2}{\hat{\beta}}. \quad (\text{E8.7})$$

The one claim spans fiscal-endowment risk if and only if

$$J_x = a(x) h_I(\alpha) J_z. \quad (\text{E8.8})$$

Adding a counterfactual second claim that completes the same externally priced shock space raises value by

$$V^{\text{complete}} - V^{\text{one claim}} = \varepsilon^2 \int_0^\infty e^{-\rho t} \frac{\|\zeta_{J,t}^\perp\|^2}{2\rho X_t^2} dt + O(\varepsilon^4). \quad (\text{E8.9})$$

The residual is the part of fiscal-wealth risk perpendicular to the claim's payoff. Equations (E8.7) and (E8.8) show exactly when it vanishes. The counterfactual second claim holds the international stochastic discount factor and deterministic allocation fixed; it isolates spanning rather than changing world equilibrium. The welfare loss is larger when residual fiscal risk is large relative to total planner wealth.

The full small-risk value has the expansion

$$V^\varepsilon = V^0 + \varepsilon^2 V^2 + O(\varepsilon^4). \quad (\text{E8.10})$$

There is no order- ε value term because mean-zero innovations have no first-order welfare effect around the deterministic allocation. Risk changes welfare at order ε^2 , while the optimal risky position remains an order-one object after the return and variance terms are scaled together. Once V^0 is known, V^2 solves a linear discounted equation with sources from the safe-rate correction, state curvature, marketed return demand, fiscal covariance, and residual unspanned risk. Exact order-two corrections to consumption and tax speed in the original nonlinear model require higher derivatives of deterministic fiscal wealth; the leading portfolio does not.

E9 Commitment, recursion, and fiscal solvency

The intended institution is date-zero commitment, while the mathematical representation is recursive. The question is whether restarting the problem later loses a promise or multiplier chosen at date zero. Under the maintained closure it does not: no private object carries an additional promise from one date to the next. Workers have no wealth or Euler equation; firms are static; domestic capital accumulation is closed by current states; external prices are exogenous and Markov; the

inherited tax rate is a state; and prior asset trades are summarized by current public net worth.

The state summary is not sufficient on its own. The same solvency and terminal convention must also apply after every history. Let ϑ be a bounded stopping time. *Restriction* takes the unchanged tail of a date-zero admissible policy and regards it as a continuation policy from S_ϑ . *Pasting* combines a feasible prefix with a measurable feasible continuation chosen from S_ϑ . Conditional no-Ponzi, terminal, and integrability requirements must survive both operations.

Conditional commitment–restart equivalence. Suppose the date-zero and recursive problems use the same future law, prices, instruments, welfare criterion, production selection, and state-based conditional solvency convention. Suppose controlled laws are well posed, conditional tails are integrable, and the admissible set is stable under stopping-time restriction and measurable pasting. Then the conditional value of an optimal date-zero tail after any reached stopping time equals the recursive value indexed by (z, x, K, N, τ) . The original optimal tail remains optimal after restart. Exact replay of the same controls additionally requires uniqueness.

The proof is the usual dynamic-programming contradiction. Restriction implies that the conditional value of the original tail cannot exceed the recursive supremum. If the recursive value were strictly greater on an event of positive probability, a measurable improving continuation could be pasted onto the original prefix and would raise date-zero welfare, a contradiction. Equality of values follows. The statement selects an optimal continuation correspondence; uniqueness is needed to select the identical control path.

The economic content is the closure: every payoff and obligation inherited from the past is represented in (z, x, K, N, τ) . Stationarity does not remove a commitment state. It is available only after the continuation set has been shown to depend on history through the current state. A promised utility, protected claim, or date-zero-only residual budget would generally add a state.

The economic solvency domain is kept schematic:

$$\mathcal{D}_{\text{sol}} = \{(z, x, k, n, \tau) : n \geq \underline{n}^{\text{sol}}(z, x, k, \tau)\}. \quad (\text{E9.1})$$

The boundary describes the continuation-viability set: states below it have no continuation satisfying future resources, instrument limits, and conditional no-Ponzi restrictions. It is not in general $N = 0$, a constant debt limit, or a claim that debt at the boundary is attainable. Better technology or a more valuable future tax base may enlarge the viable debt region; adverse states or tight policy limits may shrink it.

For a smooth boundary written in a traded log-price coordinate ℓ , diffusion tangency requires

$$\theta_b = \partial_\ell \underline{n}^{\text{sol}}. \quad (\text{E9.2})$$

Diffusion tangency says that, at the boundary, the public portfolio cannot expose net worth to an

innovation that immediately sends the state outside the feasible region. Substituting this portfolio into the drift of the boundary distance gives the remaining inward-drift inequality: expected motion must also point back into the feasible set. This is a local viability condition, not a solution for the maximal global frontier.

E10 The local quadratic benchmark

The local approximation asks how capital and the tax rate move after a small productivity or automation disturbance near the deterministic interior fixed point. It keeps the forward-looking tax trade-off but replaces the global nonlinear problem with a quadratic one. Its purpose is to generate policy derivatives and small-shock paths for comparison with the nonlinear PDE, not to replace the global problem.

Define deviations

$$y = (z - \bar{z}, x - \bar{x}, k, t)^\top, \quad k = \log(K/\bar{K}), \quad t = \tau - \frac{1}{2}. \quad (\text{E10.1})$$

The four entries have distinct roles. Productivity and automation are exogenous forces. Log capital is the slowly moving private state through which today's tax affects future wages. The tax deviation is the inherited policy state. Partition $y = (e^\top, r^\top)^\top$ with exogenous states $e = (z - \bar{z}, x - \bar{x})^\top$ and inherited policy states $r = (k, t)^\top$.

The first-order state system is

$$\dot{y} = Ay + B_\nu \nu, \quad (\text{E10.2})$$

where

$$A = \begin{pmatrix} -\kappa_z & 0 & 0 & 0 \\ 0 & -\kappa_x & 0 & 0 \\ \bar{R}/2 & \bar{R}_x/2 & -\gamma & -2\rho \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B_\nu = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad (\text{E10.3})$$

with $\gamma = (1 - \bar{\alpha})\bar{R}/2 > 0$. The first two rows are mean reversion. The third is capital accumulation: both technology states raise the net return to capital, capital abundance lowers it, and a higher tax reduces it. The final row states that tax speed ν directly controls the tax deviation.

Normalize fiscal wealth by $\bar{K}$ and write

$$\mathfrak{J}(y) = \bar{\mathfrak{J}} + j^\top y + \frac{1}{2} y^\top H^J y + o(\|y\|^2), \quad H^J = (H^J)^\top. \quad (\text{E10.4})$$

The vector j records the first-order effect of each state on fiscal wealth. The matrix H^J records value curvature and cross-effects: for example, how the value of a higher tax depends on the inherited capital stock. The local flow-value curvature is $Q^J = -Q^L$, and the value curvature is $H^J = -H^L$, relative to the loss-side convention used in the preceding LQ module. The linear

coefficient solves

$$(\rho I - A^\top)j = f_y - \tilde{\mathfrak{J}}d_r, \quad (\text{E10.5})$$

where d_r is the gradient of the zero-risk external safe rate. The curvature solves

$$0 = Q^J + A^\top H^J + H^J A - \rho H^J + \chi_{\text{LQ}} H^J B_\nu B_\nu^\top H^J, \quad \chi_{\text{LQ}} = \frac{\bar{\alpha}\kappa_\tau}{\bar{R}} > 0. \quad (\text{E10.6})$$

The economics makes this problem triangular. Technology states move on their own, then feed into capital and optimal tax adjustment; the domestic choices do not feed back into the exogenous shock laws. Write

$$A = \begin{pmatrix} A_e & 0 \\ D & A_r \end{pmatrix}, \quad A_e = -\text{diag}(\kappa_z, \kappa_x), \quad A_r = \begin{pmatrix} -\gamma & -2\rho \\ 0 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (\text{E10.7})$$

The only nonlinear equation is the 2×2 block for capital and taxes,

$$0 = Q_{rr}^J + A_r^\top H_{rr}^J + H_{rr}^J A_r - \rho H_{rr}^J + \chi_{\text{LQ}} H_{rr}^J b b^\top H_{rr}^J, \quad (\text{E10.8})$$

where

$$Q_{rr}^J = \begin{pmatrix} \rho & 2\rho \\ 2\rho & 0 \end{pmatrix}. \quad (\text{E10.9})$$

Once its stabilizing solution is selected, the technology–policy cross-effects H_{er}^J solve a linear Sylvester equation and the remaining technology curvature H_{ee}^J a discounted Lyapunov equation. This ordering mirrors the economics: first solve how the planner jointly manages capital and taxes, then ask how each exogenous shock loads onto that response.

For completeness, write $H_{rr}^J = \begin{pmatrix} h & \ell \\ \ell & m \end{pmatrix}$. The entries are curvature in capital, the capital–tax interaction, and curvature in the inherited tax rate. They solve

$$0 = \rho - (2\gamma + \rho)h + \chi_{\text{LQ}}\ell^2, \quad (\text{E10.10})$$

$$0 = 2\rho - 2\rho h - (\gamma + \rho)\ell + \chi_{\text{LQ}}\ell m, \quad (\text{E10.11})$$

$$0 = -4\rho\ell - \rho m + \chi_{\text{LQ}}m^2. \quad (\text{E10.12})$$

Let

$$a_0 = \gamma + \frac{\rho}{2}, \quad b_0 = \frac{\rho}{2}. \quad (\text{E10.13})$$

The corrected characteristic polynomial of the discounted Hamiltonian is

$$p(\lambda) = \lambda^4 - (a_0^2 + b_0^2)\lambda^2 + a_0^2 b_0^2 + 8\chi_{\text{LQ}}\rho^2\gamma. \quad (\text{E10.14})$$

For $\rho, \gamma, \chi_{\text{LQ}} > 0$, the polynomial has no imaginary-axis root. Conditional on the stated Hamil-

tonian construction, stabilizability, detectability, the second-order conditions, and selection of the stable invariant subspace, this yields the relevant real symmetric stabilizing curvature matrix. The remaining blocks are then determined when their Sylvester and Lyapunov operators are nonsingular. Absence of an imaginary root alone is not an existence theorem for the finite optimum.

The intrinsic capital–tax modes change from real to a damped complex pair when

$$16\kappa_\tau\bar{\alpha}(1 - \bar{\alpha})\rho^2 > \gamma^2(\gamma + \rho)^2. \quad (\text{E10.15})$$

When the inequality holds, tax adjustment is sufficiently responsive relative to capital’s own convergence that generic homogeneous responses contain damped oscillatory components. When it fails, the modes are real, but a capital or tax coordinate can still be nonmonotonic or overshoot because it mixes modes with different loadings and decay rates. This is a local propagation result rather than a headline finding: a nonlinear path may also hit a constraint before completing an oscillation.

The economic output of the exercise is the closed-loop transition. It supplies first-order responses of capital, taxes, wages, the tax base, and fiscal resources to productivity and automation. Adding the leading portfolio then gives first-order public-net-worth, total-planner-wealth, consumption, transfer, and financing responses. Exact precautionary corrections to consumption and tax speed are a separate higher-derivative problem.

E11 Formal scope

The appendix establishes exact static allocation results; exact interior HJB and portfolio identities; deterministic fiscal-wealth and tax results under the stated transversality condition; local automation derivatives; small-risk projection and order-two welfare formulas; conditional commitment equivalence; and a conditional characterization of the local stabilizing quadratic solution. The results belong to different scopes: the firm allocation and interior portfolio identity are exact within their stated branches; the portfolio-finance formulas are leading small-risk results; and the capital–tax transition is local to the deterministic fixed point.

It does not establish a globally viable state space, existence or uniqueness of the constrained five-state value, satisfaction of terminal conditions by an optimal feedback, a solved maximal solvency frontier, or a stationary distribution. Nor does it establish that the policy generated by the nonlinear problem remains in the smooth production and policy region from every admissible initial state. Those are mathematical and computational tasks for the nonlinear implementation.

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